Aidan Moriarty

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EDUCATION

Washington University in St. Louis

Master of Science, Aerospace Engineering; GPA: 4.0 — May 2027
Bachelor of Science, Mechanical Engineering; GPA: 3.71 — May 2026
DII Men's Club Rugby — 2024 - Present

Oklahoma City University

Bachelor of Science, Pre-Engineering, Minor: Mathematics; GPA: 3.7 — May 2026
DI Men's Varsity Rowing — 2021 - 2024

STUDENT EXPERIENCE

Chief Engineer, Mission Operations Subteam | WashU Satellite — Feb. 2026 - Present

- Lead a 10-member multidisciplinary subteam through mission design for all future satellite and high-altitude balloon missions, setting schedules, assigning tasks, allocating resources, and providing technical oversight across 4 concurrent projects.
- Own licensing for SCALAR, the club's first CubeSat, managing the FCC application and IARU frequency coordination to keep the mission compliant with regulatory requirements and on track for a 2027 launch.
- Run feasibility assessments that confirm mission viability, derive requirements and constraints, and give design feedback to other subteams that validates and improves their design choices.
- Authored the club's mission operations standards, including proposal, Concept of Operations, flight operations procedure, and analysis-of-alternatives templates modeled on NASA, ESA, and University Nanosatellite Program standards.
- Designed the subteam's onboarding project, in which new members write a CubeSat mission proposal to grow a radish on-orbit using a template that teaches mission design fundamentals, getting new members contributing 3 weeks after joining.

Responsible Engineer, SCALAR Payload Submodule | WashU Satellite — Jan. 2026 - Present

- Own design, manufacture, integration, and testing of the SCALAR payload, a system of magnetorquer rods and a control algorithm providing 3-axis attitude control on the club's first CubeSat using only Earth's magnetic field, advancing technology for future CubeSat missions.
- Designed a precision wire tensioner in SOLIDWORKS that enabled 4x more wire turns per torque rod with consistent, tightly packed layers, improving magnetic field uniformity and manufacturability.
- Created a MATLAB magnetorquer design tool that maximizes magnetic dipole moment under CubeSat mass, power, and volume constraints, enabling trade studies of design alternatives and giving future missions a reusable selection tool.

PROFESSIONAL EXPERIENCE

Operations Analyst Intern | Modern Technology Solutions, Inc. — May 2026 - August 2026

- Modeled close collaborative aircraft platforms for operator-in-the-loop AFSIM simulations by creating decision trees integrated with situational awareness to demonstrate autonomous capabilities.
- Created a python tool that parses radar cross section plots by mapping the location of each pixel on the plot relative to the origin, which could then be used on simulated platforms.
- Assisted in the development of a tool that takes data from OpenStreetMap and creates route networks in AFSIM replicating real routes, enabling realistic simulation of how those route networks can be utilized.

RESEARCH EXPERIENCE

Undergraduate Researcher | WashU Scalable Integrated MicroSystems Lab — Jan. 2026 - May 2026

- Conducted research investigating the use of microfluidic devices for studying carbonation reactions and critical element recovery from low-grade mineral deposits.
- Simulated fluid flow through variable-sized bead packings in a weir geometry using COMSOL for comparison to experimental results.
- Prepared a final research summary of key results and methods used during time in lab and provided insights into future applications of microsystem mineral recovery.

SKILLS

Engineering Software: SOLIDWORKS, COMSOL, WarpWare, AFSIM, Orbital STK
Amateur Radio: Technician Class License, Callsign KE9GJR